

Final Exam

● Graded

Student

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Total Points

91 / 100 pts

Question 1

Question 1

16 / 18 pts

1.1 a

8 / 10 pts

– 0 pts Correct

– 2 pts first is wrong

– 2 pts second is wrong

✓ – 2 pts third is wrong

– 2 pts fourth is wrong

– 2 pts fifth is wrong

1.2 b

8 / 8 pts

✓ – 0 pts Correct

– 4 pts sigma is positive

– 4 pts Q is negative

– 8 pts not done or wrong

– 3 pts explanation is wrong

Question 2

Question 2

16 / 16 pts

2.1 a

8 / 8 pts

✓ - 0 pts Correct

- 3 pts COP is wrong

- 2 pts Kelvin scale is not used

- 8 pts not done or wrong

- 3 pts COP equation is wrong

2.2 b

8 / 8 pts

✓ - 0 pts Correct

- 3 pts $h_2 = 2833$

- 2 pts Equation is wrong

- 8 pts not done or wrong

- 2 pts $h_1 = 3192$

- 2 pts $h_1 - h_2 = 100$

Question 3

Question 3

27 / 32 pts

3.1 a 2 / 2 pts

✓ - 0 pts Correct

3.2 b 3 / 3 pts

✓ - 0 pts Correct

- 1 pt closed system

3.3 c 4 / 4 pts

✓ - 0 pts Correct

- 2 pts energy balance is wrong

- 2 pts entropy balance is wrong

3.4 d 8 / 8 pts

✓ - 0 pts Correct

- 1 pt $v_2=0.4015$

- 1 pt $P=2.7$ bar

- 1 pt unit conversion is wrong

- 2 pts need to multiply with mass

- 1 pt u_2 is 1742

- 1 pt $v_1=1.069 \times 10^{-3}$

- 1 pt $u_1=546.02$

- 4 pts Q is not calculated or equated to W . (Q is not equal to work in this example)

- 8 pts not done or wrong

- 3 pts work equation is wrong

3.5 e 5 / 5 pts

✓ - 0 pts Correct

- 1 pt $s_2=4.86$

- 1 pt $s_1=1.634$

- 2 pts need to multiply with mass

- 5 pts not done or wrong

3.6 f

5 / 5 pts

✓ - 0 pts Correct

- 1 pt $T = (20 + 273)K$

- 1 pt calculation error

- 3 pts equation is wrong

- 5 pts not done or wrong

3.7 g

0 / 5 pts

- 0 pts Correct

✓ - 5 pts not done or wrong

- 1 pt s2 on TS is wrong

- 2 pts s1 in both are wrong

- 1 pt s1 in Pv wrong

- 1 pt s2 on Pv wrong

Question 4

Question 4

32 / 34 pts

4.1 — a

3 / 4 pts

– 0 pts Correct

– 1 pt steady state

✓ – 1 pt control volume

– 4 pts not done or wrong

4.2 — b

6 / 6 pts

✓ – 0 pts Correct

– 6 pts not done

4.3 — c

5 / 6 pts

– 0 pts Correct

– 6 pts not done or wrong

– 3 pts equation is wrong

✓ – 1 pt calculation error

4.4 — d

6 / 6 pts

✓ – 0 pts Correct

– 3 pts can not use Temperatures

– 6 pts not done or wrong

– 1 pt calculation error

4.5 — e

6 / 6 pts

✓ – 0 pts Correct

– 1 pt calculation error

– 3 pts equation is wrong

– 6 pts not done or wrong

✓ - 0 pts Correct

- 1 pt calculation error

- 6 pts not done or wrong

- 3 pts s1 s2 not calculated

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (18 pts)

True or False (2 pts each)

F Energy transferred by heat to a closed system during an irreversible system can be calculated by the area under the curve in T-v diagram.

T Entropy, S , is extensive property.

T An ideal gas in a piston-cylinder assembly expands isothermally, doing work and receiving an equivalent amount of energy by heat transfer from the surrounding atmosphere. This process of the gas is in violation of the Kelvin-Planck statement of the second law.

F The maximum thermal efficiency of *any* power cycle operating between hot and cold reservoirs at 1000°C and 500°C, respectively, is 50%.

T The change in entropy of a closed system is the same for every process between two specified end states.

Fill in the blank

(8 pts) A closed system undergoes an internally irreversible process in which work is done on the system and the heat transfer occurs from the system only at a location on the boundary where the temperature is T_b . Determine whether the entropy change of the system is positive, negative, zero, or indeterminate. Explain.

$$\Delta S = \int \frac{\delta q}{T_b} + \sigma$$

Heat transfer from system: $q < 0$
 process internally irreversible: $\sigma > 0$

So $\Delta S = \frac{Q}{T_b} + S_{gen}$, since there are no values for Q or σ , it is currently impossible to determine the sign of entropy change, indeterminate

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QUESTION 2 (16 pts)

- a) (8 pts) An inventor has developed a refrigerator capable of maintaining its freezer compartment at 20°F while operating in a kitchen at 70°F, and claims the device has a coefficient of performance of 10. Evaluate the claim.

$$T_L = 20^\circ\text{F} = 479.67 \text{ R}$$

$$T_H = 70^\circ\text{F} = 529.67 \text{ R}$$

$$\text{COP}_R = 10$$

$$\text{COP}_R = \frac{T_L}{T_H - T_L} = \frac{479.67}{529.67 - 479.67} = 9.59$$

The inventor said the COP_R was 10, while I calculated the max COP_R was 9.59. I believe the inventor was false.

- b) (8 pts) Water vapor enters a turbine at 3 bar $T_1 = 360^\circ\text{C}$ and exits at $p_2 = 0.7$ bar. The work developed is measured as 100 kJ per kg of water vapor flowing. Calculate the isentropic turbine efficiency.

$$P_1 = 3 \text{ bar}$$

$$P_2 = 0.7 \text{ bar}$$

$$T_1 = 360^\circ\text{C} = 633.15 \text{ K}$$

$$W = 100 \text{ kJ/kg}$$

$$h_1 (3 \text{ bar}, 360^\circ\text{C}) = 3192.2 \text{ kJ/kg}$$

$$s_1 (3 \text{ bar}, 360^\circ\text{C}) = 7.9061 \text{ kJ/kg}\cdot\text{K}$$

$$\eta_t = \frac{W}{W_{is}} = \frac{h_1 - h_2}{h_1 - h_{2s}} = \frac{100}{358.6} = 0.279$$

$$W_{is} = h_1 - h_{2s} = 3192.2 - 2833.6 = 358.6 \text{ kJ/kg}$$

$$\eta_t = 0.279$$

$$\text{At } P_2 = 0.7$$

$$T = 160^\circ\text{C}: h_{2,160} = 2798.2 \text{ kJ/kg}$$

$$s_{2,160} = 7.8279 \text{ kJ/kg}\cdot\text{K}$$

$$T = 200^\circ\text{C}: h_{2,200} = 2876.7 \text{ kJ/kg}$$

$$s_{2,200} = 8.0012 \text{ kJ/kg}\cdot\text{K}$$

Interpolate:

$$\frac{7.9061 - 7.8279}{8.0012 - 7.8279} = 0.451 = F$$

$$h_{2s} = 7.9061 \text{ kJ/kg}$$

$$h_{2s} = h_{160} + F(h_{200} - h_{160}) = 2798.2 + 0.451(2876.7 - 2798.2) = 2833.6 \text{ kJ/kg}$$

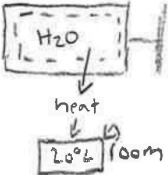
$$W = h_1 - h_2 = 100 \text{ kJ/kg}$$

$$h_2 = h_1 - 100$$

QUESTION 3: (32 pts)

5 kg of water, initially a saturated liquid at 130°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 20°C room until the water becomes a state where $x=0.6$. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (3 pts) State the engineering model
- (4 pts) Write the simplified energy balance and simplified entropy balance.
- (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
- (5 pts) Determine the change in entropy of the water during the process, **ΔS , in kJ/K**
- (5 pts) Determine the entropy generated during the process, **σ , in kJ/K**. State if this process is reversible, irreversible or impossible.
- (5 pts) Carefully sketch the process on P-v and T-s diagram.

a) 

b) Closed system
 $\Delta KE = \Delta PE = 0$
 Constant pressure
 sat liquid $\rightarrow$ sat mixture

c) $\Delta U = q - w$
 $m(u_2 - u_1) = q - w$
 $\Delta S = \frac{Q}{T_b} + \sigma$
 $m(s_2 - s_1) = \frac{Q}{T_b} + \sigma$

d) $v_2 = v_f + x(v_g - v_f)$
 $= 0.0010697 + 0.6(0.6685 - 0.0010697)$
 $= 0.402 \text{ m}^3/\text{kg}$
 $V_2 = m v_2 = 5(0.402) = 2.01 \text{ m}^3$
 $u_2 = u_f + x(u_g - u_f)$
 $= 546.02 + 0.6(2539.9 - 546.02)$
 $= 1742.35 \text{ kJ/kg}$
 $s_2 = s_f + x(s_g - s_f)$
 $= 1.6144 + 0.6(7.0269 - 1.6144)$
 $= 4.87 \text{ kJ/kg} \cdot \text{K}$

$v_1 = v_f = 0.0010697 \text{ m}^3/\text{kg}$
 $V_1 = m v_1 = 5(0.0010697) = 0.00535 \text{ m}^3$
 $u_1 = u_f = 546.02 \text{ kJ/kg}$
 $s_1 = s_f = 1.6144 \text{ kJ/kg} \cdot \text{K}$

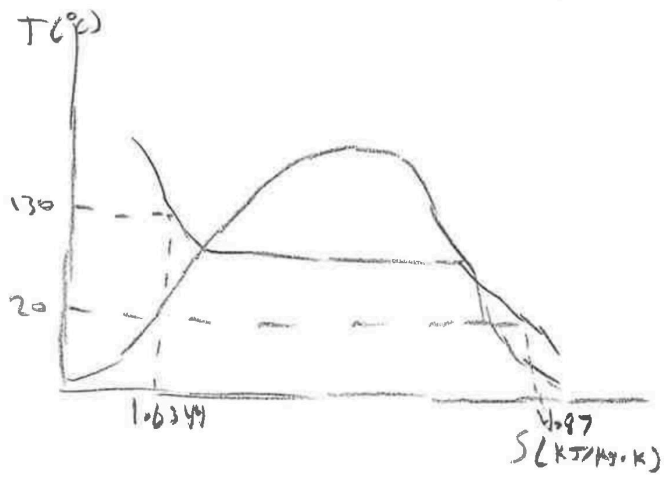
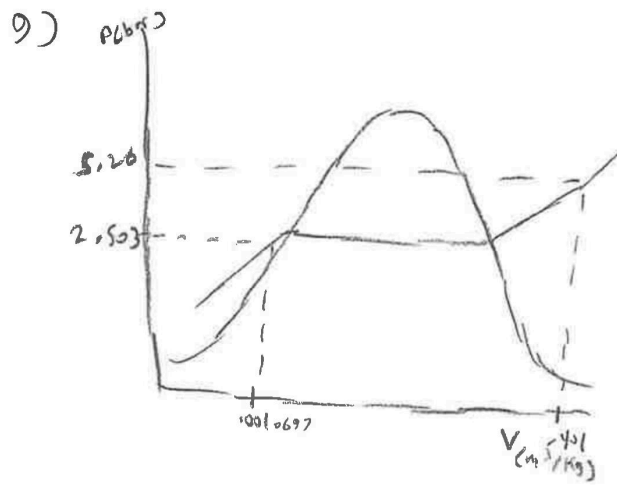
$w = P(V_2 - V_1)$
 $= 270(2.01 - 0.00535)$
 $= 541.26 \text{ kJ}$

$Q = \Delta U + w = m(u_2 - u_1) + w$
 $= 5(1742.35 - 546.02) + 541.26$
 $= 6522.91 \text{ kJ}$

e) $\Delta S = m(s_2 - s_1)$
 $= 5(4.87 - 1.6144)$
 $= 16.178 \text{ kJ/kg} \cdot \text{K}$

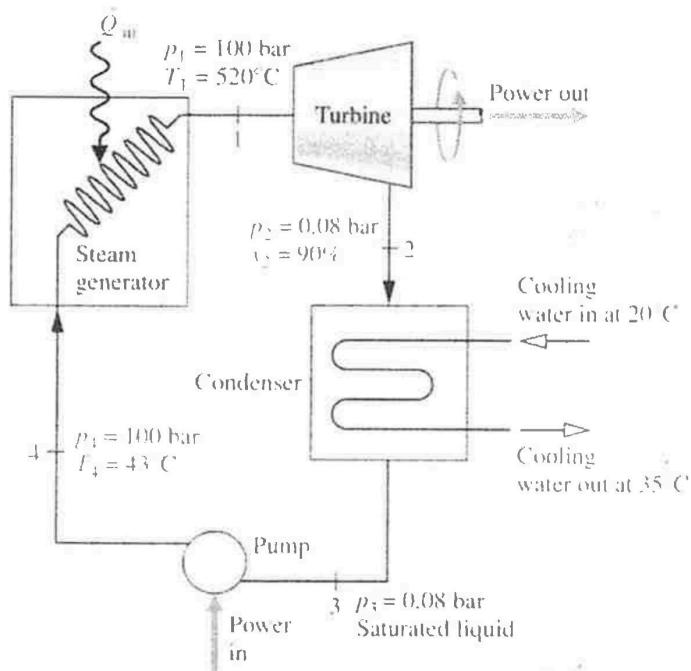
f) $\Delta S = \frac{Q}{T_b} + \sigma$
 $\sigma = \Delta S - \frac{Q}{T_b}$
 $= 16.178 - \frac{6522.91}{293}$
 $= -7.715 \text{ kJ/K}$
 This process is impossible

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QUESTION 4: (34 pts)

The figure below shows a simple vapor power plant operating at steady state with water as the working fluid. Data at key locations are given on the figure. The mass flow rate of the water circulating through the components is 109 kg/s. Stray heat transfer and kinetic and potential energy effects can be ignored.



- (4pts) State the engineering model.
- (6pts) Write simplified balances for mass, energy and entropy.
- (6pts) Calculate the net power developed, in MW.
- (6pts) Calculate the thermal efficiency.
- (6pts) Determine the mass flow rate of the cooling water, in kg/s.
- (6pts) Determine the rates of entropy production, in kW/K, for the turbine.

a) 1 inlet, 1 outlet: turbine, condenser, pump, steam generator

$$\Delta KE = \Delta PE = 0$$

water

Turbine: adiabatic, shaft work output
pump: adiabatic, shaft work input
steady state

entropy)

$$\text{Turbine: } \dot{S}_{\text{gen}} = \dot{m}(s_2 - s_1)$$

$$\text{Condenser: } \dot{S}_{\text{gen}} = \dot{m}(s_3 - s_2) + \frac{\dot{Q}_0}{T_{\text{cw}}}$$

b) ^{mass}
$$\frac{dm_{\text{cv}}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$$

$$\dot{m}_i = \dot{m}_e$$

Energy)

$$\frac{dE_{\text{cv}}}{dt} = \dot{Q}_{\text{cv}} - \dot{W}_{\text{cv}} + \sum \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

$$\text{Turbine: } \dot{W}_t = \dot{m}(h_1 - h_2)$$

$$\text{Condenser: } \dot{Q}_0 = \dot{m}(h_2 - h_3)$$

$$\text{Pump: } \dot{W}_p = \dot{m}(h_4 - h_3)$$

$$\text{Steam: } \dot{Q}_1 = \dot{m}(h_1 - h_4)$$

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c) state 1: $P_1 = 100 \text{ bar}$, $T_1 = 520^\circ\text{C}$

$$h_1 = 3425.1 \text{ kJ/kg}$$

$$s_1 = 6.6622 \text{ kJ/kg}\cdot\text{K}$$

state 2: $P_2 = .08 \text{ bar}$, $x_2 = .9$

$$h_f = 173.88 \text{ kJ/kg}$$

$$h_{fg} = 2403.1 \text{ kJ/kg}$$

$$s_f = .5926 \text{ kJ/kg}\cdot\text{K}$$

$$s_{fg} = 8.2287 \text{ kJ/kg}\cdot\text{K}$$

$$h_2 = h_f + x(h_{fg}) = 173.88 + .9(2403.1) = 2336.67 \text{ kJ/kg}$$

$$s_2 = s_f + x(s_{fg}) = .5926 + .9(8.2287 - .5926) = 7.465 \text{ kJ/kg}\cdot\text{K}$$

state 3: sat liquid at .08 bar

$$h_3 = h_f = 173.88 \text{ kJ/kg}$$

$$v_3 = v_f = .0010084 \text{ m}^3/\text{kg}$$

$$\text{Turbine work: } w_t = h_1 - h_2 = 3425.1 - 2336.67 = 1088.43 \text{ kJ/kg}$$

$$\text{Pump work: } w_p = v_3(P_4 - P_3)$$

$$= .0010084(100,000 - 8) = 10.08 \text{ kJ/kg}$$

$$h_4 = h_3 + w_p = 173.88 + 10.08 = 183.96 \text{ kJ/kg}$$

$$w_{\text{net}} = w_t - w_p = 1088.43 - 10.08 = 1078.35 \text{ kJ/kg}$$

$$\dot{W}_{\text{net}} = \dot{m} w_{\text{net}} = 109(1078.35) = 117540 \text{ kW} = 117.54 \text{ MW}$$

d) $q_{\text{in}} = h_1 - h_4 = 3425.1 - 183.96 = 3241.14 \text{ kJ/kg}$

$$\eta = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{1078.35}{3241.14} = .333 \quad \eta = 33.4\%$$

e) $\dot{m}_{\text{cw}} = \frac{\dot{m} q_{\text{out}}}{c_p(T_{\text{out}} - T_{\text{in}})}$

$$= \frac{109(2162.79)}{4.18(15)}$$

$$= 3759.87 \text{ kg/s}$$

$$q_{\text{out}} = h_2 - h_3 = 2336.67 - 173.88 = 2162.79 \text{ kJ/kg}$$

f) $\dot{S}_{\text{gen}} = \dot{m}(s_2 - s_1)$

$$= 109(7.465 - 6.6622)$$

$$= 87.51 \text{ kW/K}$$

